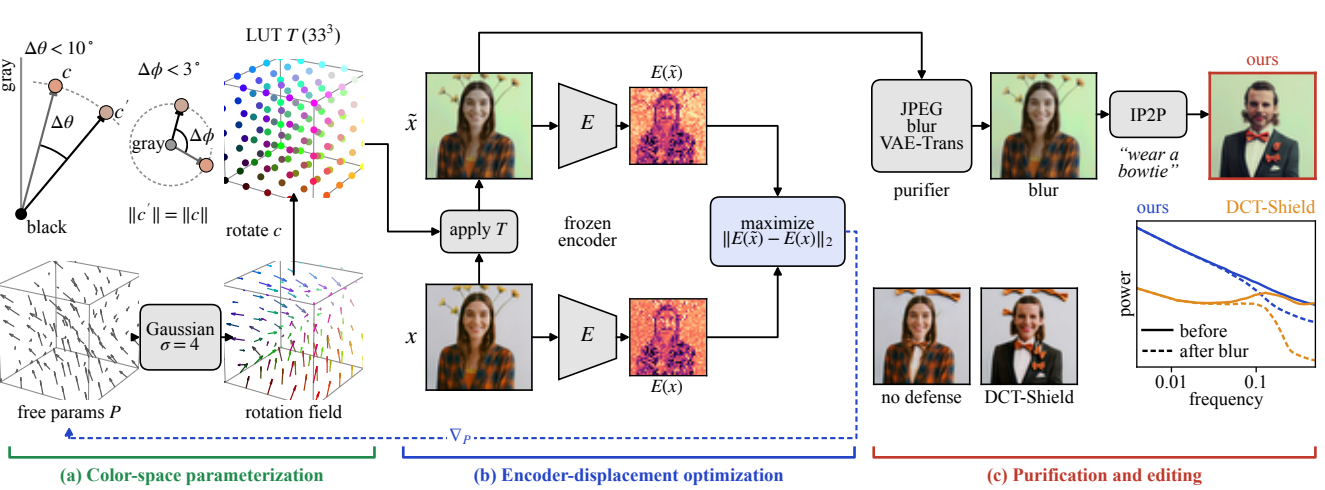

$\Delta\theta < 10^\circ$
gray
c
c'
$\Delta\theta$
black
$\Delta\phi < 3^\circ$
gray
$\Delta\phi$
$\|c'\| = \|c\|$
LUT T (33^3)
rotate c
free params P
Gaussian $\sigma = 4$
rotation field
$\tilde{x}$
apply T
x
E
E
$E(\tilde{x})$
$E(x)$
frozen encoder
maximize $\|E(\tilde{x}) - E(x)\|_2$
∇_P
JPEG blur VAE-Trans
purifier
blur
IP2P
"wear a bowtie"
ours
no defense
DCT-Shield
ours
DCT-Shield
power
before
after blur
0.01
0.1
frequency
(a) Color-space parameterization
(b) Encoder-displacement optimization
(c) Purification and editing